

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2082 Kartik

Exam.		Back	
Level	BE	Full Marks	80
Programme	BEI	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Communication Systems (EX 656)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

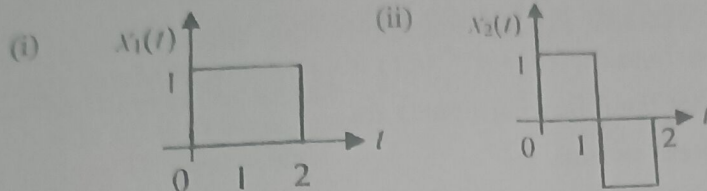
1. Describe the functional digital communication system working in a half-duplex mode and contrast it with an analog communication system. Mention all the communication impairments and performance parameters. [5+3]
2. Define the power spectral density function and express its relation with the autocorrelation function. Check whether $y(t)=\log\{x(t)\}$ is a linear or non-linear system. [5+5]
3. Taking examples of any two signals, derive auto-correlation, cross-correlation and convolution graphically. [3+3+3]
4. a) Draw the timing diagram of Unipolar RZ, Polar NRZ, AMI and B8ZS for a binary sequence of 101100001000000000. [6]
b) Discuss the limitations of Shannon Hartley Channel capacity theorem. [2]
5. Derive the expression of a double-tone AM and draw its line spectrums. Consider a sinusoidal signal $m(t) = A \cos \omega_m t$ applied to a delta modulator with step size Δ . Show that the slope overload distortion will occur if $> \frac{\Delta}{\omega_m T_s}$. [5+3]
6. With suitable examples, introduce how CRC-3 and LBC work to correct a single bit error. [5+5]
7. Check whether the received code word 110111 as has error or not. Encode "Rain Rain Go To Spain" using Shannon-Fano encoder and calculate the transmission efficiency. Find the codeword if the message is received as "Rain Rain Go Go Spain". [7+2+3]
8. Write short notes/solutions on: (Any Three) [3×5]
 - a) Constellation diagram of a 32-QAM
 - b) Impulse response of a matched filter for an AWGN channel
 - c) A (7,4) polynomial cyclic Hamming code given by $g(x) = 1 + x + x^3$
 - d) Single bit error correction using Viterbi decoder
 - e) Packet switching vs message switching

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1. Define BW, SNR, SQNR, BER, ISI parameters. [5]
2. a) Perform convolution of the following signals, by graphical method. [6]



- b) Construct the signal flow diagram of the system described by the following equation. [4]

$$y(t) + 2 \int y(t) dt + 3 \iint y(t) dt dt = 4 \int x(t) dt + 5 \iint x(t) dt dt.$$

3. Encode 100100000111 using Unipolar RZ, AMI and HDB3 coding schemes. [6]
4. a) Prepare a constellation diagram of 32-QAM. [4]
- b) Illustrate MSK signal for a message of 1011010. [5]
5. With the help of necessary mathematical expressions of Bandwidth and Modulation Index, differentiate between AM and FM. [6]
6. With the help of neat circuit diagram, explain the operation of the square law detector. [6]
7. With suitable assumption for a code word of $C(7,4)$, demonstrate how Hamming Code detects an error. [8]
8. With a suitable example of input bit stream show State Diagram, Code Tree and Trellis Diagram. Demonstrate how a convolutional code corrects a single error. [6+4]
9. Encode 'Phoola Ko Thunga Bahera Gayo Ganga Ko Panima' using Shannon-Fano coding technique and calculate the transmission efficiency. [6+2]
10. Write short notes on: (Any Three) [3x4]
 - a) Hilbert transform and its properties
 - b) Delta modulation
 - c) T1 vs E1 hierarchy
 - d) Band limited white noise vs thermal noise
 - e) Multiplexing vs multiple access

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1. Draw the generic block diagram of Analog communication system for half-duplex mode. Explain the need and disadvantages of modulation. [3+5]
2. Define energy and power signals with example. Write the properties of LTI system. [4+4]
3. Calculate the percentage of power saved in: (i) DSB-SC (ii) SSB-SC as compared to standard AM if the modulation depth is 60%. Describe QAM modulation and prepare constellation diagram of 16-QAM. [2+2+4]
4. Explain the aim of line coding. Encode "Ma Mare Pani Mero Desh Bachi Rahos" using Shannon-Fano codes and calculate its efficiency. [10]
5. Why is non-uniform quantization required? Show that for a voice transmission, the basic data rate using PCM is 64 kbps. [3+3+4]
6. State and explain Shannon-Hartley channel capacity theorem with its implications. Represent 100111010 using Unipolar RZ, Bipolar NRZ, Manchester and AMI encoding technique. [4+6]
7. Explain Pre-amphasis and De-amphasis. [4+4]
8. Briefly explain filter and oscillator requirement in Frequency Division Multiplexing (FDM). Compare packet switching and message switching. [4+4]
9. Construct a (7, 4) binary CRC using a generator polynomial $g(x) = x^3 + x^2 + 1$ with data Vector (1011). Demonstrate how CRC-4 detects two burst errors. [5+5]

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Subject: - Communication System I (EX 652)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
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1. What is communication system? Explain the importance of transducers, modulators and channel in Analog communication system with neat and clean diagram. [2+6]
2. What is distortionless transmission? Differentiate energy and power signal. Describe the impulse response of LPF. [2+3+3]
3. What is autocorrelation and cross correlation of a signal? Explain and derive power spectrum density function of a signal. [2+4]
4. What is conventional amplitude modulation? Explain the ring modulator for the generation of double side band amplitude modulation with neat and clear diagram. [2+6]
5. Calculate the power saving for double side band AM signal and single side band AM signal for percentage modulation of 60%. [6]
6. What is superheterodyne AM receiver? Explain the coherent detection and phase error effect of DSB AM. [2+4]
7. What is PLL? Explain the generation of Narrow Band FM using Armstrong method. Why amplitude limiter is used in limiter discriminator method? [2+4+2]
8. Differentiate NBFM and WBFM? Explain the functioning of stereo encoder and decoder in commercial FM with clear diagram. [3+5]
9. A modulated signal is given by $s(t) = 40 \cos(100.1 \times 10^6 \times 2\pi t + 10 \sin(15 \times 10^3 \times 2\pi t))$. Determine (i) practical bandwidth (ii) frequency sensitivity (iii) frequency deviation and d) modulation index. [8]
10. Explain the FDM telephone hierarchy. Describe the filter and oscillator requirement in FDM. [3+3]
[2+4]
11. Write short notes on:
 - a) ASK, PSK and FSK
 - b) Properties of Bessel function